

Topic 15: Linear Systems of ODEs & Phase Plane Portraits

Professor Leonard Master Series

Course: Applied Ordinary Differential Equations (ENGR 213 Companion)

1. Whiteboard Intuition: Coupled Dynamics in the Phase Plane

Professor Leonard explains the matrix formulation of differential equations:

"When multiple physical quantities interact (like two connected springs or predators and prey), you don't have just one derivative—you have a coupled vector system $\mathbf{x}' = \mathbf{A}\mathbf{x}$. The eigenvalues λ of matrix $\mathbf{A}$ are the growth/oscillation rates, and the eigenvectors $\mathbf{v}$ are the invariant directions in space! Solutions march along these straight-line eigen-tracks!"

$$\det(\mathbf{A} - \lambda\mathbf{I}) = 0$$

$$\mathbf{x}(t) = c_1 e^{\lambda_1 t} \mathbf{v}_1 + c_2 e^{\lambda_2 t} \mathbf{v}_2$$

2. Professor Leonard's Whiteboard Problem Walkthroughs

Problem 15.1: Real Distinct Eigenvalues & Saddle Point

Statement: Solve the system $\mathbf{x}' = \begin{pmatrix} 1 & 2 & 3 \\ 2 & 1 & -2 \end{pmatrix} \mathbf{x}$. Classify the equilibrium at the origin.

Step-by-Step Whiteboard Solution:

- Step 1: Characteristic Equation** $\det(\mathbf{A} - \lambda\mathbf{I}) = 0$:

$$\begin{vmatrix} 1 - \lambda & 2 & 3 \\ 2 & 1 - \lambda \end{vmatrix} = (1 - \lambda)(1 - \lambda) - 6 = \lambda^2 - 2\lambda - 4 = 0$$

$$(\lambda - 4)(\lambda + 1) = 0 \implies \lambda_1 = 4, \quad \lambda_2 = -1$$

- Step 2: Find Eigenvector for $\lambda_1 = 4$:** $(\mathbf{A} - 4\mathbf{I})\mathbf{v} = \mathbf{0}$

$$\begin{pmatrix} -3 & 2 & 3 \\ 2 & -3 & -2 \end{pmatrix} \begin{pmatrix} v_1 \\ v_2 \\ v_3 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}$$

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$$-3v_1 + 2v_2 + 3v_3 = 0 \implies 2v_2 + 3v_3 = 3v_1 \implies \mathbf{v}_1 = \begin{pmatrix} 2 \\ 3 \\ 1 \end{pmatrix}$$

- Step 3: Find Eigenvector for $\lambda_2 = -1$:** $(\mathbf{A} - (-1)\mathbf{I})\mathbf{v} = \mathbf{0}$

$$\begin{pmatrix} 2 & 2 & 3 \\ 2 & 2 & 3 \end{pmatrix} \begin{pmatrix} v_1 \\ v_2 \\ v_3 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}$$

$$\begin{pmatrix} 2 & 1 \\ 0 & 0 \end{pmatrix} \begin{pmatrix} v_1 \\ v_2 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}$$

$$2v_1 + v_2 = 0 \implies v_2 = -2v_1 \implies \mathbf{v}_2 = \begin{pmatrix} 1 \\ -2 \end{pmatrix}$$

- **Step 4: General Solution Vector:**

$$\mathbf{x}(t) = c_1 e^{4t} \begin{pmatrix} 2 \\ 3 \end{pmatrix} + c_2 e^{-t} \begin{pmatrix} 1 \\ -1 \end{pmatrix}$$

- **Step 5: Phase Portrait Classification:**

- $\lambda_1 = +4 > 0$: Unstable direction along line $y = \frac{3}{2}x$.
- $\lambda_2 = -1 < 0$: Stable direction along line $y = -x$. Opposite signs $\implies$ **SADDLE POINT (Unstable)**.

Problem 15.2: Complex Conjugate Eigenvalues & Spiral Sink

Statement: Solve $\mathbf{x}' = \begin{pmatrix} -1 & -4 \\ 1 & -1 \end{pmatrix} \mathbf{x}$.

Step-by-Step Whiteboard Solution:

- **Step 1: Characteristic Equation:**

$$\det(\mathbf{A} - \lambda \mathbf{I}) = (-1 - \lambda)^2 - (-4) = (\lambda + 1)^2 + 4 = 0 \implies \lambda = -1 \pm 2i$$

- **Step 2: Stability:** Real part $\text{Re}(\lambda) = -1 < 0 \implies$ amplitudes decay exponentially e^{-t} . Imaginary part $\text{Im}(\lambda) = 2 \implies$ rotates with period π . **SPIRAL SINK (Asymptotically Stable Attractor)**.

Problem 15.3: Converting 2nd-Order ODE to a First-Order System

Statement: Convert $y'' + 4y' + 3y = 0$ into a 2×2 first-order linear system.

Step-by-Step Whiteboard Solution:

- **Step 1: Define state variables:** Let $x_1 = y$ and $x_2 = y'$.
- **Step 2: Differentiate state variables:**

$$x_1' = y' = x_2$$

$$x_2' = y'' = -3y - 4y' = -3x_1 - 4x_2$$

- **Step 3: Matrix form:**

$$\begin{pmatrix} x_1' \\ x_2' \end{pmatrix} = \begin{pmatrix} 0 & 1 \\ -3 & -4 \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \end{pmatrix}$$

3. Common Exam Traps & Professor Leonard Warnings

- **Trap 1: Saddle Point Stability:** A saddle point is NEVER stable. Even though one direction decays ($e^{-t} \rightarrow 0$), almost all trajectories eventually get pulled into the expanding e^{4t} mode.

- **Trap 2: Eigenvector Free Parameter:** If $-3v_1 + 2v_2 = 0$, you can choose $v_1 = 2, v_2 = 3$. Any non-zero multiple works!